

10(a)	<i>any 9 of:</i> <i>structure</i> 1 microvilli ; A brush border 2 many mitochondria ; 3 tight junctions ; 4 folded, basal membrane / described ; 5 many, transport proteins / cotransporters / pumps / carrier proteins ; <i>process</i> 6 active transport of Na^+, out of (pct) cells / into blood ; 7 Na^+ ion concentration, gradient (produced) / reduced in cell ; 8 Na^+ enters (pct) cells from, lumen / tubule / filtrate ; 9 by facilitated diffusion / using carrier protein ; 10 cotransport (from lumen to pct cell) ; 11 of, glucose / amino acids / ions ; <i>only award if mp10 given</i> 12 AVP ; e.g. water enters (pct) cells	9
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Question	Answer	Marks
10(b)	<i>any 6 of:</i> 1 hypothalamus detects water potential of blood ; 2 (using) osmoreceptors ; 3 (if) water potential, decreases / is low ; 4 ADH produced ; 5 <i>ref. to</i> neurosecretion / neurosecretory cells ; 6 which extend into posterior pituitary ; 7 ADH released from posterior pituitary ; 8 enters blood stream ; 9 water potential of blood, increases / returns to set point ; 10 negative feedback ;	6